

MPM Risk Register Takeaway: This is a full, standards-aligned **risk register** for the Mechanical Power Module (MPM), structured for engineering, logistics, safety, interoperability, and consortium-level deployment. Each risk includes a clear description, impact, likelihood, mitigation, and monitoring pathway. Guided Links are embedded so you can expand any category.

Risk Register Overview

The MPM introduces a universal mechanical-power primitive, so its risks span **mechanical engineering, interface integrity, safety envelope, logistics, host-system variability, economic adoption, and consortium governance**. This register is designed for pilot deployment, standards refinement, and multi-vendor scaling.

Engineering & Interface Risks

- **Mechanical Misalignment** — *Impact:* High. Misalignment can cause spline wear, vibration, or unsafe coupling. *Likelihood:* Medium during early pilots. *Mitigation:* Alignment guides, operator training, metrology checks. *Monitoring:* Weekly interface-tolerance measurements.
- **Spline Wear Acceleration** — *Impact:* Medium–High. Reduces module lifespan and host compatibility. *Likelihood:* Medium. *Mitigation:* Hardened materials, lubrication standards, wear-rate telemetry. *Monitoring:* μm -level wear tracking per 100 operating hours.
- **Latch/Interlock Failure** — *Impact:* Very High. Could cause uncontrolled disconnect. *Likelihood:* Low with redundancy. *Mitigation:* Dual mechanical + electrical interlocks, emergency stop logic. *Monitoring:* 99.98% reliability threshold.

Safety Envelope Risks

- **Uncontrolled Disconnect** — *Impact:* Critical. Zero-tolerance safety event. *Likelihood:* Very Low if compliant. *Mitigation:* Redundant retention, interlock sequencing, operator certification. *Monitoring:* Real-time interlock telemetry.
- **Thermal Overrun** — *Impact:* Medium. Can degrade interface or trigger safety shutdown. *Likelihood:* Medium under heavy load. *Mitigation:* Thermal sensors, cooling envelope, load-profile limits. *Monitoring:* Temperature rise $<15^{\circ}\text{C}$ over baseline.
- **Vibration Envelope Exceedance** — *Impact:* Medium. Causes wear and misalignment. *Likelihood:* Medium. *Mitigation:* Vibration dampers, host-system balancing. *Monitoring:* $<4\text{ mm/s}$ RMS threshold.

Logistics & Swap-Out Risks

- **Swap-Out Time Variability** — *Impact:* Medium. Affects operational uptime and logistics planning. *Likelihood:* High early in deployment. *Mitigation:* Standardized swap-station layout, operator training. *Monitoring:* Target <15 minutes (95th percentile).
- **Transport Shock Exposure** — *Impact:* Medium. Can damage interface or internal components. *Likelihood:* Medium. *Mitigation:* Shock-rated transport frames, handling protocols. *Monitoring:* g-force telemetry during transport.
- **Storage Environment Drift** — *Impact:* Low–Medium. Environmental drift affects seals and tolerances. *Likelihood:* Medium. *Mitigation:* Environmental envelope standards. *Monitoring:* Humidity/temperature logs.

Host-System Interoperability Risks

- **Host-Side Variability** — *Impact:* High. Non-standard hosts may cause misalignment or unsafe coupling. *Likelihood:* High across diverse industries. *Mitigation:* Modular adapters, compliance testing. *Monitoring:* Host-compatibility reports per deployment.
- **Adapter Wear** — *Impact:* Medium. Affects long-term interoperability. *Likelihood:* Medium. *Mitigation:* Hardened materials, periodic replacement. *Monitoring:* µm-level wear tracking.
- **Host-Side Safety Events** — *Impact:* Medium–High. Host systems may not meet MPM safety envelope. *Likelihood:* Medium. *Mitigation:* Host-side safety audit, adapter-level interlocks. *Monitoring:* Event logging.

Telemetry & Data Risks

- **Telemetry Loss or Corruption** — *Impact:* High. Prevents compliance verification. *Likelihood:* Medium. *Mitigation:* Redundant data paths, buffered logging. *Monitoring:* Telemetry uptime >99.5%.
- **Sensor Calibration Drift** — *Impact:* Medium. Causes inaccurate compliance data. *Likelihood:* Medium. *Mitigation:* NIST-traceable calibration schedule. *Monitoring:* Monthly calibration checks.

Economic & Adoption Risks

- **Lifecycle Cost Uncertainty** — *Impact:* Medium–High. Affects procurement decisions. *Likelihood:* Medium during early pilots. *Mitigation:* Pilot-based cost modeling, multi-vendor sourcing. *Monitoring:* Quarterly economic analysis.
- **Vendor Lock-In Risk** — *Impact:* High if standard not enforced. *Likelihood:* Medium. *Mitigation:* Consortium governance, compliance certification. *Monitoring:* Vendor interoperability metrics.



Consortium & Standards Risks

- **Standards Fragmentation** — *Impact:* High. Undermines universality. *Likelihood:* Medium without strong governance. *Mitigation:* Consortium charter, compliance enforcement. *Monitoring:* Annual standards review.
- **Compliance Drift** — *Impact:* Medium–High. Vendors may deviate from tolerances. *Likelihood:* Medium. *Mitigation:* Certification workflow, periodic audits. *Monitoring:* Compliance reports per vendor.



Monitoring Cadence

- **Continuous:** telemetry, interlock state, mechanical performance
- **Weekly:** interface wear, safety envelope review
- **Monthly:** logistics performance, host-system compatibility
- **Quarterly:** economic analysis, standards refinement
- **Annually:** consortium compliance audit